



SCEGGS Darlinghurst

2009

HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 2

This is a TRIAL PAPER only and does not necessarily reflect the content or format of the Higher School Certificate Examination for this subject.

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1–8
- All questions are of equal value

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Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (15 marks)

(a) Find $\int_0^{\frac{\pi}{3}} \sec^3 x \tan x \, dx$ 2

(b) Find $\int \sqrt{\frac{5-x}{5+x}} \, dx$. 3

(c) (i) Find real numbers A , B and C such that 2

$$\frac{10}{(3+x)(1+x^2)} \equiv \frac{A}{3+x} + \frac{Bx+C}{1+x^2}.$$

(ii) Use the substitution $t = \tan \theta$ to find $\int \frac{10}{3 + \tan \theta} \, d\theta$. 3

(d) For $n \geq 1$, let $I_n = \int_0^1 \frac{dx}{(x^2 + 1)^n}$.

(i) By writing $\int_0^1 \frac{dx}{(x^2 + 1)^n}$ as $\int_0^1 1 \times \frac{dx}{(x^2 + 1)^n}$, and using integration by parts, 3
 show that

$$2nI_{n+1} = 2^{-n} + (2n-1)I_n.$$

(ii) Hence evaluate $\int_0^1 \frac{dx}{(x^2 + 1)^3}$. 2

End of Question 1

Question 2 (15 marks) Use a SEPARATE writing booklet.

- (a) The complex number w is given by $w = -1 + \sqrt{3}i$.
- (i) Show that $w^2 = 2\bar{w}$. 1
- (ii) Evaluate $|w|$ and $\arg w$. 2
- (iii) Show that w is a root of $z^3 - 8 = 0$. 1
- (b) On separate diagrams, draw a neat sketch of the locus defined by
- (i) $|z - 1 - 3i| \leq 2$ and $\frac{\pi}{4} \leq \arg z \leq \frac{\pi}{2}$. 2
- (ii) $\arg\left(\frac{z-i}{z-1}\right) = \frac{\pi}{2}$. 2
- (c) By considering the binomial expansion of $(1+i)^n$ show that 3
- $$1 - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \dots = 2^{\frac{n}{2}} \cos \frac{n\pi}{4}.$$
- (d) The points O, I, Z and P on the Argand Plane represent the complex numbers $0, 1, z$ and $z + 1$ respectively, where $z = \cos \theta + i \sin \theta$ is any complex number of modulus 1, with $0 < \theta < \pi$.
- (i) Explain why $OIPZ$ is a rhombus. 1
- (ii) Show that $\frac{z-1}{z+1}$ is purely imaginary. 2
- (iii) Find the modulus of $z + 1$ in terms of θ . 1

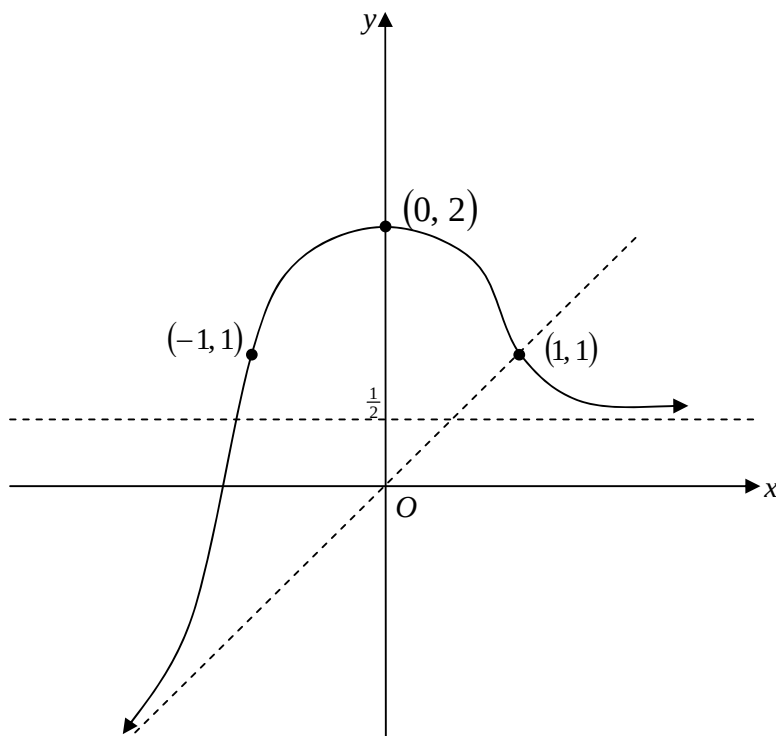
End of Question 2

Question 3 (15 marks) Use a SEPARATE writing booklet.

- (a) The locus defined by $|z - 2| - |z + 2| = 2$ corresponds to part of a hyperbola 3
in the Argand Plane.

Sketch the locus labeling the foci, directrices, asymptotes and any intercepts with the axes.

(b)



The diagram shows the graph of $y = f(x)$. The lines $y = x$ and $y = \frac{1}{2}$ are both asymptotes.

On the answer page provided, draw separate sketches of the following graphs.

Clearly indicate any important features.

(i) $y = (f(x))^2$ 2

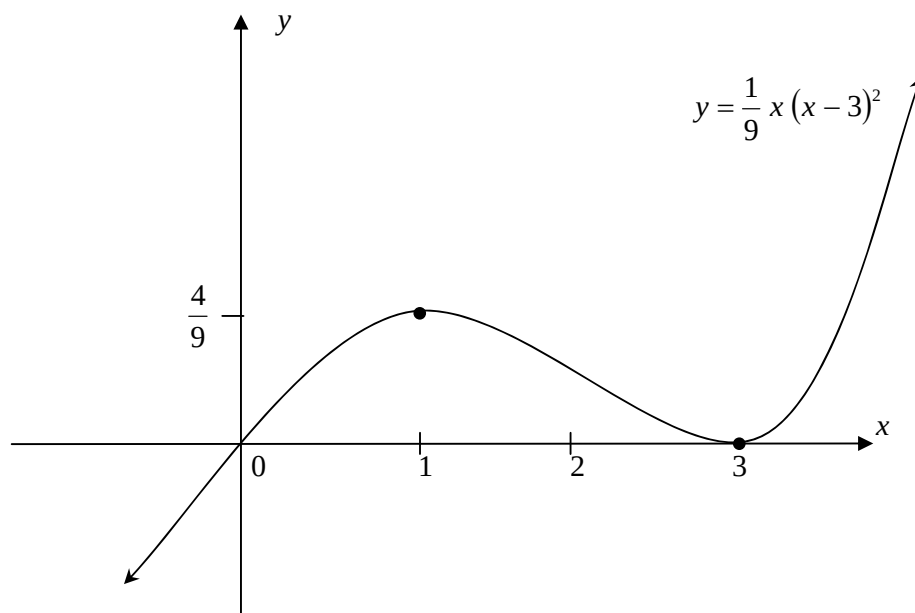
(ii) $y = \frac{1}{f(x)}$ 2

(iii) $y = f(x) - x$ 2

Question 3 continues on page 5

Question 3 (continued)

(c)



- (i) Given the sketch of $y = \frac{1}{9} x (x - 3)^2$ above, sketch the curve 2

$$y^2 = \frac{1}{9} x (x - 3)^2.$$

- (ii) Use implicit differentiation to find $\frac{dy}{dx}$ in terms of x and y for 1

$$y^2 = \frac{1}{9} x (x - 3)^2.$$

- (iii) Given that the length of a curve between $x = a$ and $x = b$ is given by 3

$$\int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx,$$

find the entire length of the curve $y^2 = \frac{1}{9} x (x - 3)^2$ for $0 \leq x \leq 3$.

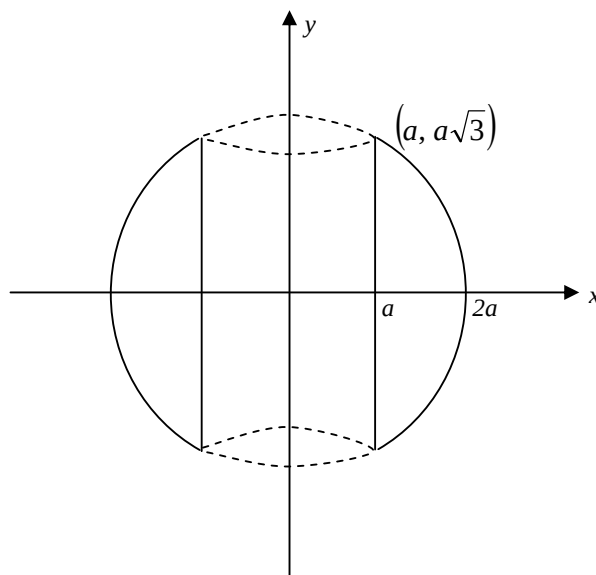
End of Question 3

Question 4 (15 marks) Use a SEPARATE writing booklet.

- (a) Consider the polynomial $P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$.
- (i) If $P(x)$ has zeroes $a + bi$ and $a + 2bi$, where a and b are real and $b > 0$, find the values of a and b . 3
- (ii) Hence express $P(x)$ as the product of two quadratic factors with real coefficients. 1
- (b) The region bounded by the curve $y = \cos x$ and the coordinate axes is rotated about the y -axis. 3

Use the method of cylindrical shells to find the volume of the solid formed.

- (c) 3



A cylindrical hole of radius a cm is bored through the centre of a sphere of radius $2a$ cm.

Show that the volume of the remaining solid is $4\sqrt{3}a^3\pi$ cm³.

Question 4 continues on page 7

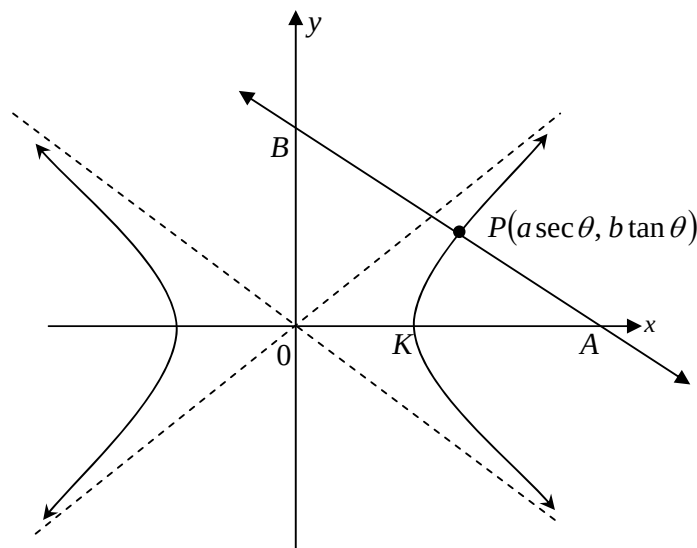
Question 4 (continued)

- (d) The cubic equation $x^3 + kx + 1 = 0$, where k is a constant, has roots α , β and γ .
For each positive integer n , $S_n = \alpha^n + \beta^n + \gamma^n$.
- (i) State the value of S_1 and express S_2 in terms of k . 2
- (ii) Show that for all n , $S_{n+3} + kS_{n+1} + S_n = 0$. 2
- (iii) Hence, or otherwise, express $\alpha^4 + \beta^4 + \gamma^4$ in terms of k . 1

End of Question 4

Question 5 (15 marks) Use a SEPARATE writing booklet.

(a)



The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where $a > b > 0$, cuts the positive x -axis at the point K .

The normal to the hyperbola at the point $P(a \sec \theta, b \tan \theta)$ cuts the x -axis at A and the y -axis at B , as shown in the diagram.

- (i) Show that the equation of the normal to the hyperbola at the point P is 2

$$\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2.$$

- (ii) Find the midpoint M of AB . 3

- (iii) Find the point G such that G divides the interval OM in the ratio 2:1. 1

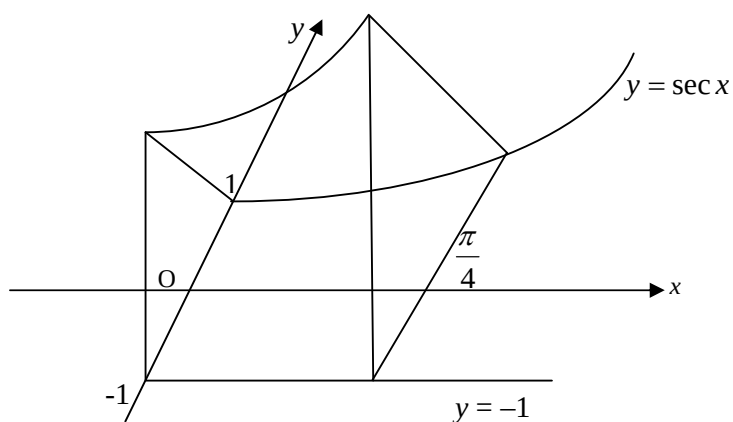
- (iv) Show that the locus of G is a hyperbola and find the point L at which this locus cuts the positive x -axis. 3

- (v) If $\frac{OL}{OK} < 1$, show that $1 < e < \sqrt{3}$. 2

Question 5 continues on page 9

Question 5 (continued)

- (b) The base of a solid is the region in the xy plane enclosed by the curve $y = \sec x$ and $y = -1$ for $0 \leq x \leq \frac{\pi}{4}$. Each cross-section perpendicular to the x -axis is an equilateral triangle.



- (i) Show that the area of the triangular cross-section at a distance x from the y -axis is $\frac{\sqrt{3}}{4} (\sec x + 1)^2$. 1
- (ii) Hence find the volume of the solid. 3

End of Question 5

Question 6 (15 marks) Use a SEPARATE writing booklet.

(a) Let $I_1 = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$ and let $I_2 = \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx$

(i) Using a suitable substitution show that $I_1 = I_2$. 1

(ii) Find the value of $I_1 + I_2$ and hence evaluate $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$. 3

(b) Let $z = \cos \theta + i \sin \theta$ be any complex number of modulus 1.

(i) Show that $\frac{z^2 - 1}{z} = 2i \sin \theta$. 2

(ii) Using the formula for the sum of a Geometric Progression and the result in part (i), prove that 2

$$z + z^3 + z^5 + z^7 + z^9 = \frac{\sin 10\theta + i(1 - \cos 10\theta)}{2 \sin \theta}.$$

(iii) Hence write down a simplified expression for 3

$$\cos \theta + \cos 3\theta + \cos 5\theta + \cos 7\theta + \cos 9\theta$$

and find the general solution to

$$\cos \theta + \cos 3\theta + \cos 5\theta + \cos 7\theta + \cos 9\theta = \frac{1}{2}.$$

Question 6 continues on page 11

Question 6 (continued)

- (c) Seven players are entered in a round robin tennis competition. Each round consists of three singles matches with the 7th player obtaining a bye.

In how many ways can the first round of the competition be arranged if

- | | | |
|------|----------------------------|---|
| (i) | there are no restrictions? | 2 |
| (ii) | Amy is not playing Ben? | 2 |

End of Question 6

Question 7 (15 marks) Use a SEPARATE writing booklet.

(a) Suppose $x > 0$, $y > 0$, $z > 0$.

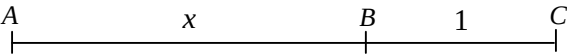
(i) Prove that $x^2 + y^2 \geq 2xy$. 1

(ii) Hence, or otherwise, prove that $\frac{x}{y} + \frac{y}{z} \geq 2$. 1

(iii) Prove that $x^3 + y^3 \geq xyz \left(\frac{x}{z} + \frac{y}{z} \right)$. 1

(iv) Hence show that $x^3 + y^3 + z^3 \geq 3xyz$. 1

(v) Deduce that $(a+b+c)(a+b+d)(a+c+d)(b+c+d) \geq 81abcd$ where $a > 0$, $b > 0$, $c > 0$, $d > 0$. 1

(b) (i)  1

The diagram shows a straight line segment AC divided by B in the ratio $x:1$.

If A divides CB externally in the same ratio that B divides AC internally, show that

$$x^2 = x + 1$$

(ii) A sequence $\{F_n\}$, the Fibonacci numbers, is defined by $F_1 = 1$, $F_2 = 1$ and $F_{n+1} = F_n + F_{n-1}$ for $n \geq 2$. 3

The golden ratio φ , and its conjugate root θ , are the positive and negative solutions to the equation in part (i).

Prove by induction, that the closed form expression for the Fibonacci numbers is given by

$$F_n = \frac{\varphi^n - \theta^n}{\sqrt{5}}.$$

Question 7 continues on page 13

Question 7 (continued)

- (c) A projectile is fired vertically upwards with initial speed u . It experiences air resistance proportional to its speed as well as gravitational acceleration g , so that in its upwards flight, the equation of motion is $\ddot{x} = -g - kv$, for some constant $k > 0$ and where v is the velocity of the projectile.

- (i) Show that the time T taken to reach its maximum height is given by 3

$$T = \frac{1}{k} \log_e \left(1 + \frac{ku}{g} \right).$$

- (ii) By first writing $\ddot{x}$ as $v \frac{dv}{dx}$, show that the maximum height of the particle H 3
is given by

$$H = \frac{u - gT}{k}.$$

End of Question 7

- (ii) Hence show that $n^n c^{n-2} = 4b^n (n-2)^{n-2}$. **2**

(v) By making further use of similar triangles, or otherwise, show that $AK = KH$. 3

page 14

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

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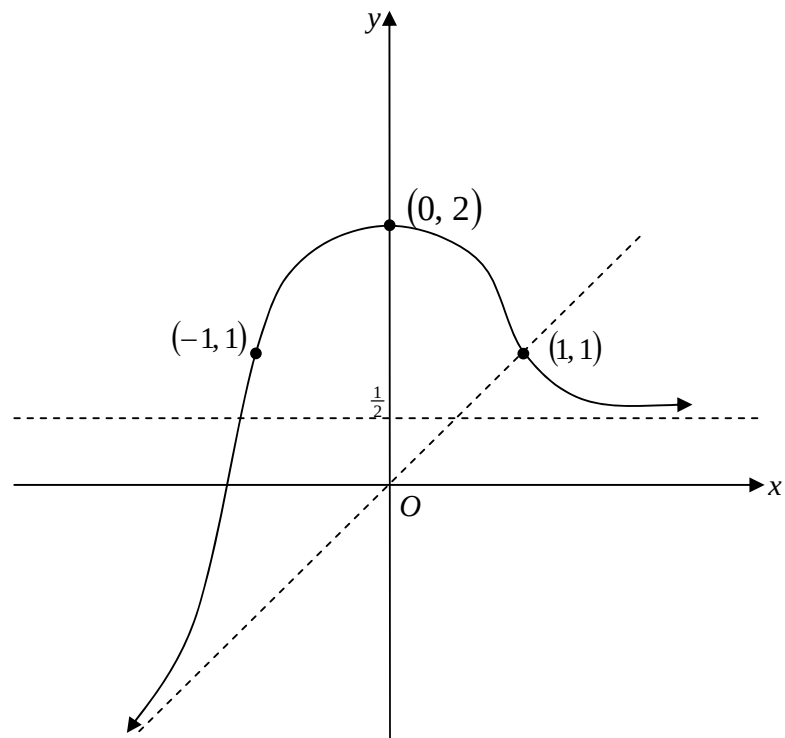
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Questions 3 (b)

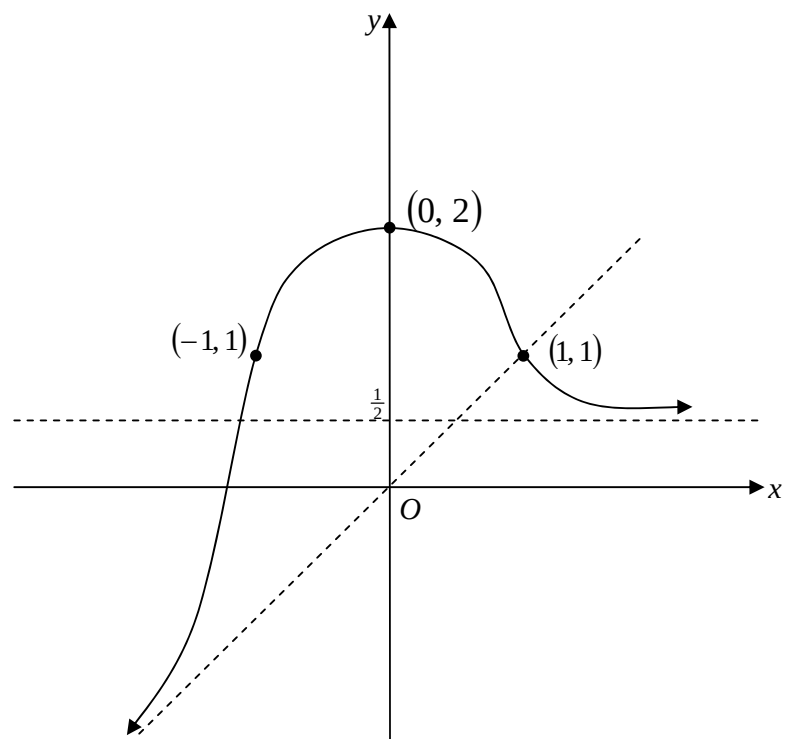
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(i)

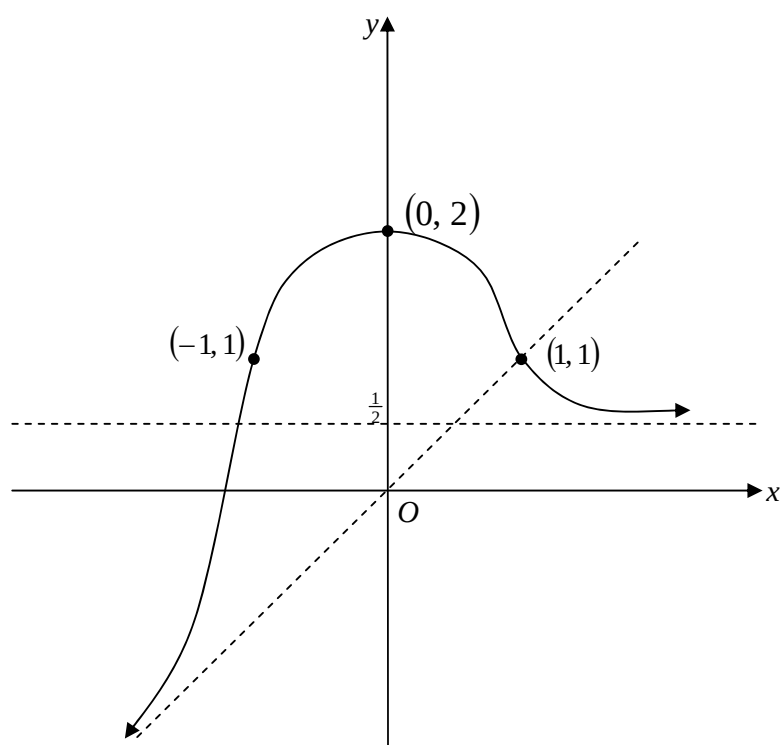


(ii)



Question 3b (continued)

(iii)



Question 1 (15 marks)

Calc 15

$$\begin{aligned}
 (a) \quad & \int_0^{\pi/3} \sec^3 x \tan x \, dx \\
 &= \int_0^{\pi/3} \sec^2 x \cdot \sec x \tan x \, dx \\
 &= \left[\frac{1}{3} \sec^3 x \right]_0^{\pi/3} = \frac{7}{3} \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad & \int \sqrt{\frac{5-x}{5+x}} \, dx \\
 &= \int \frac{5-x}{\sqrt{25-x^2}} \, dx \quad \left. \begin{array}{l} \text{rationalise} \\ \text{the numerator} \end{array} \right\} \\
 &= \int \frac{5}{\sqrt{25-x^2}} + \frac{1}{2} \int \frac{-2x}{\sqrt{25-x^2}} \, dx \checkmark \\
 &= 5 \sin^{-1}\left(\frac{x}{5}\right) + \sqrt{25-x^2} + C \checkmark
 \end{aligned}$$

$$(c) (i) \quad \frac{10}{(3+x)(1+x^2)} = \frac{\boxed{1}}{3+x} + \frac{\boxed{-1}x + \boxed{3}}{1+x^2}$$

$$A=1, B=-1, C=3 \quad \checkmark \checkmark$$

$$(ii) \quad t = \tan \theta$$

$$\frac{dt}{d\theta} = \sec^2 \theta = 1 + \tan^2 \theta$$

$$d\theta = \frac{dt}{1+t^2}$$

$$\therefore \int \frac{10}{3+\tan \theta} \, d\theta$$

$$= \int \frac{10}{(3+t)(1+t^2)} \, dt \checkmark$$

Working for
substitution
must be shown.

Overall, I don't think Q1 was as well done as it should have been. Mistakes ranged from trivial + & - signs to much more fundamental ones. You should be aiming for 15/15 in 15 minutes for this question.

This question did not require the use of t-formulae!

$$\begin{aligned}
&= \int \frac{1}{3+t} + \frac{-t+3}{1+t^2} dt \\
&= \int \frac{1}{3+t} - \frac{1}{2} \int \frac{2t}{1+t^2} + \int \frac{3}{1+t^2} dt \\
&= \ln(3+t) - \frac{1}{2} \ln(1+t^2) + 3 \tan^{-1}(t) + C \\
&= \ln \left(\frac{3+\tan \theta}{\sqrt{1+\tan^2 \theta}} \right) + 3\theta + C \\
&= \ln(3\cos \theta + \sin \theta) + 3\theta + C
\end{aligned}$$

In this particular question, no marks were deducted for not changing back to θ , but don't forget it because they usually are (deducted)!

$$\begin{aligned}
\text{(d) (i) } I_n &= \int_0^1 x \frac{dx}{(x^2+1)^n} \\
&= \left[x \cdot \frac{1}{(x^2+1)^n} \right]_0^1 - \int_0^1 x \cdot \frac{-n \cdot 2x}{(x^2+1)^{n+1}} dx \\
&= \frac{1}{2^n} + 2n \int_0^1 \frac{x^2+1}{(x^2+1)^{n+1}} - \frac{1}{(x^2+1)^{n+1}} dx \\
&= 2^{-n} + 2n(I_n - I_{n+1})
\end{aligned}$$

+1-1 to not change the question is a technique worth putting in your toolbox.

$$2n I_{n+1} = 2^{-n} + (2n-1) I_n$$

$$\text{(ii) } I_1 = \int_0^1 \frac{dx}{x^2+1} = [\tan^{-1} x]_0^1 = \frac{\pi}{4}$$

$$2I_2 = \frac{1}{2} + I_1 = \frac{1}{2} + \frac{\pi}{4}$$

$$I_2 = \frac{1}{4} + \frac{\pi}{8}$$

$$4I_3 = \frac{1}{4} + 3I_2 = \frac{1}{4} + \frac{3}{4} + \frac{3\pi}{8}$$

$$I_3 = \frac{1}{4} + \frac{3\pi}{32}$$

This part can certainly be completed whether or not you could complete (i).

Question 2 (15 marks)

Comm 7

(a) $w = -1 + \sqrt{3}i$

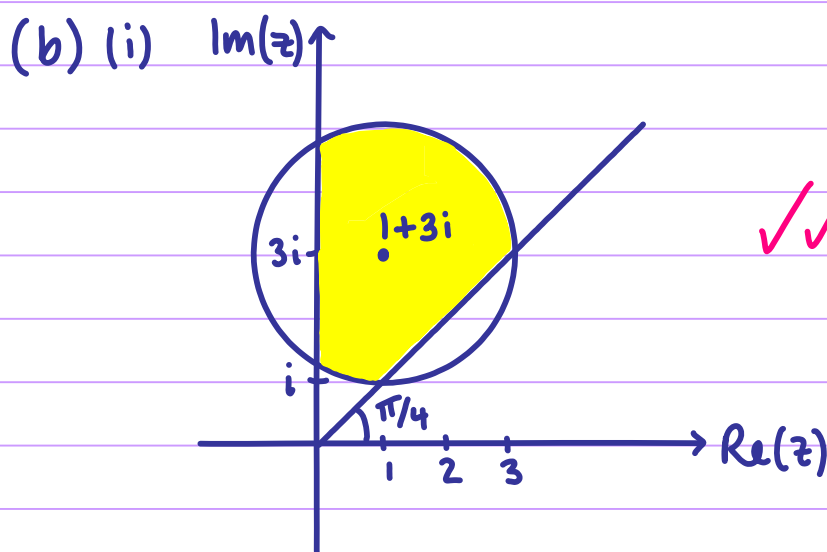
(i) $w^2 = (-1 + \sqrt{3}i)^2$
 $= 1 - 2\sqrt{3}i - 3$
 $= -2 - 2\sqrt{3}i$
 $= 2(-1 - \sqrt{3}i)$
 $= 2\bar{w}$ ✓

(ii) $|w| = 2$ ✓
 $\arg w = \frac{2\pi}{3}$ ✓

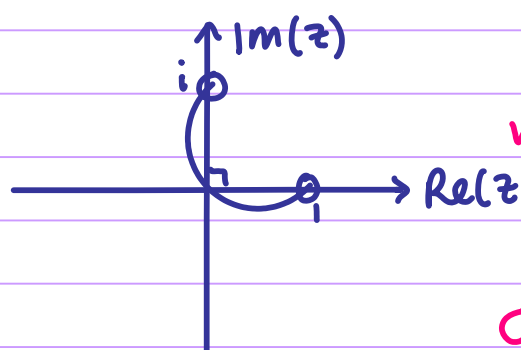
(iii) $w^3 - 8 = (2\text{cis}\frac{2\pi}{3})^3 - 8$
 $= 2^3 \text{cis } 2\pi - 8$
 $= 8 \times 1 - 8$
 $= 0$

∴ w is a root of $z^3 - 8 = 0$ ✓

While there was the option of finding all three roots & showing w was one of them (using mod-arg or cartesian form), this was not the fastest approach!



(ii) $\arg \left(\frac{z-i}{z-1} \right) = \frac{\pi}{2}$



Comm 4

You must draw these graphs as carefully & as accurately & as to scale as you can. If you can't draw a circle invest in a compass & whether or not you think you can draw a straight line use a ruler!

Notice the open circles at i & 1 and that the locus passes through the origin.

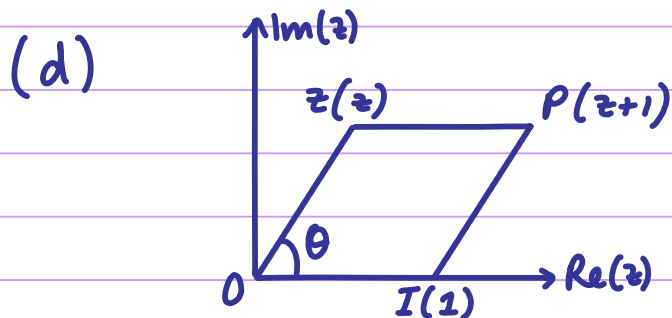
$$(c) (1+i)^n = \binom{n}{0} + \binom{n}{1}i + \binom{n}{2}i^2 + \binom{n}{3}i^3 + \binom{n}{4}i^4 + \dots$$

$$(\sqrt{2} \operatorname{cis} \pi/4)^n = \left[\binom{n}{0} - \binom{n}{2} + \binom{n}{4} - \dots \right] + i \left[\binom{n}{1} - \binom{n}{3} + \binom{n}{5} - \dots \right]$$

$$2^{n/2} \operatorname{cis} \frac{n\pi}{4} = \left[\binom{n}{0} - \binom{n}{2} + \binom{n}{4} - \dots \right] + i \left[\binom{n}{1} - \binom{n}{3} + \binom{n}{5} - \dots \right] \quad \checkmark \checkmark$$

Taking the real part of both sides

$$2^{n/2} \cos \frac{n\pi}{4} = 1 - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \dots \quad \checkmark$$



$$(i) \vec{OP} = z+1 = \vec{OI} + \vec{OZ}$$

$\therefore OIPZ$ is a parallelogram
Since $|OZ| = |OI| = 1$, the adjacent sides of $OIPZ$ are equal & thus $OIPZ$ is a rhombus. $\checkmark$

Not many could clearly explain this obvious enough fact.

Comm 1

$$(ii) \arg \left(\frac{z-1}{z+1} \right) = \text{angle of rotation from } (z+1) \text{ to } (z-1) \\ = \text{angle of rotation from } \vec{OP} \text{ to } \vec{IZ} \\ = \pi/2 \text{ (diagonals of a rhombus are } \perp)$$

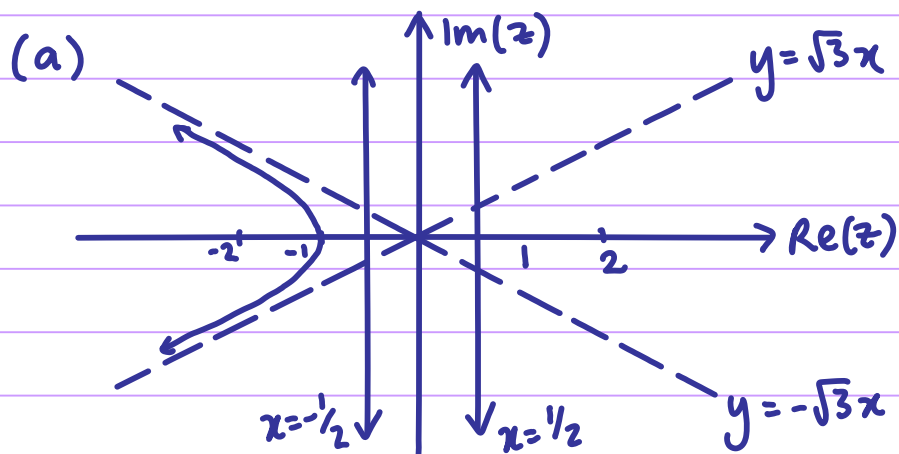
You have to explain why $\arg \left(\frac{z-1}{z+1} \right)$ is the angle between the diagonals.

$$\therefore \frac{z-1}{z+1} \text{ is purely imaginary } \checkmark \checkmark$$

Comm 2

$$(iii) |z+1| = \sqrt{1^2 + 1^2 - 2 \times 1 \times 1 \times \cos(\pi - \theta)} \quad \text{OR} \quad \cos \theta/2 = \frac{|z+1|/2}{1} \\ = \sqrt{2 + 2 \cos \theta} \quad [\text{cos rule}] \quad \checkmark \quad |z+1| = 2 \cos \theta/2 \quad [\text{SOHCAHTOA}]$$

Question 3 (15 marks)



$$\begin{aligned}\text{Foci : } (\pm 2, 0) &\Rightarrow ae = 2 \\ 2a = 2 &\Rightarrow a = 1 \\ \therefore e = 2 \text{ \& } b^2 &= a^2(e^2 - 1) \\ b &= \sqrt{3}\end{aligned}$$

$$\text{Directrices : } x = \pm \frac{1}{2}$$

$$\text{Asymptotes : } y = \pm \sqrt{3}x$$

$$x\text{-intercepts : } x = \pm 1$$

$$\text{Note : } |z - 2| - |z + 2| = 2$$

$\Rightarrow$ distance from z to 2 is greater than the distance from z to -2.

✓ correct branch
✓✓ correct features

Comm 3

Calc 4, Comm 11

This is the easiest version of this question you could possibly get — and it wasn't too successful.

The question even told you it was only a branch of the hyperbola!

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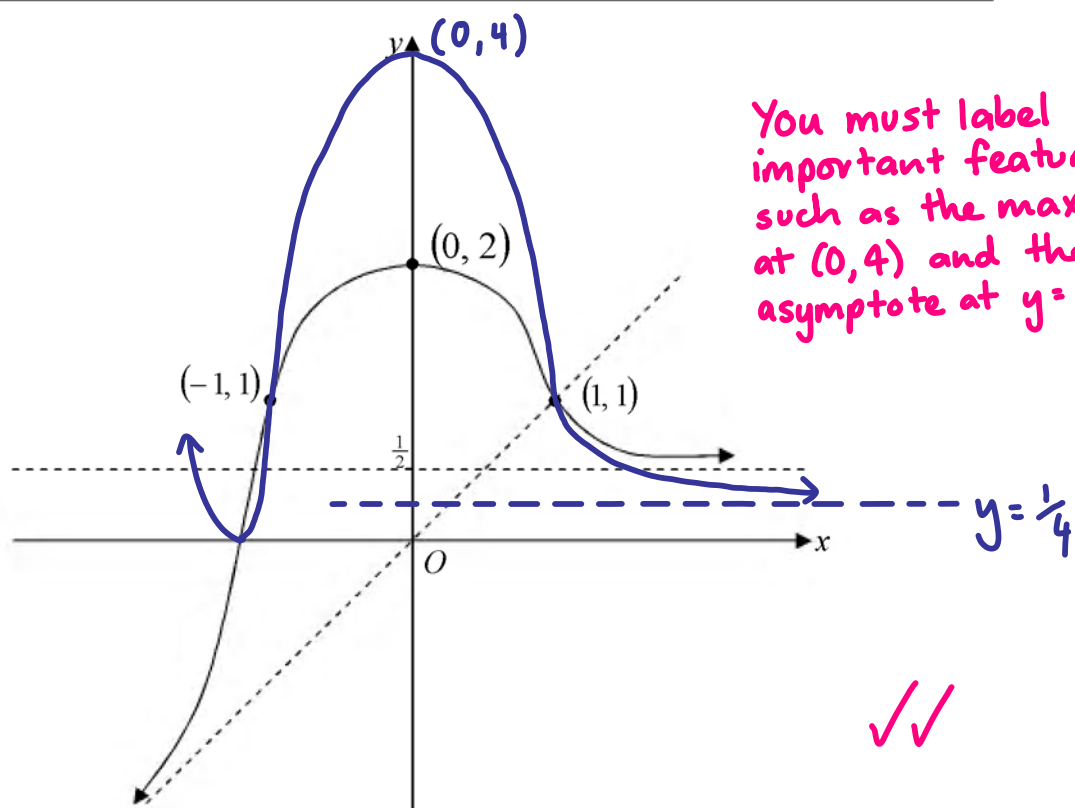
Questions 3 (b)

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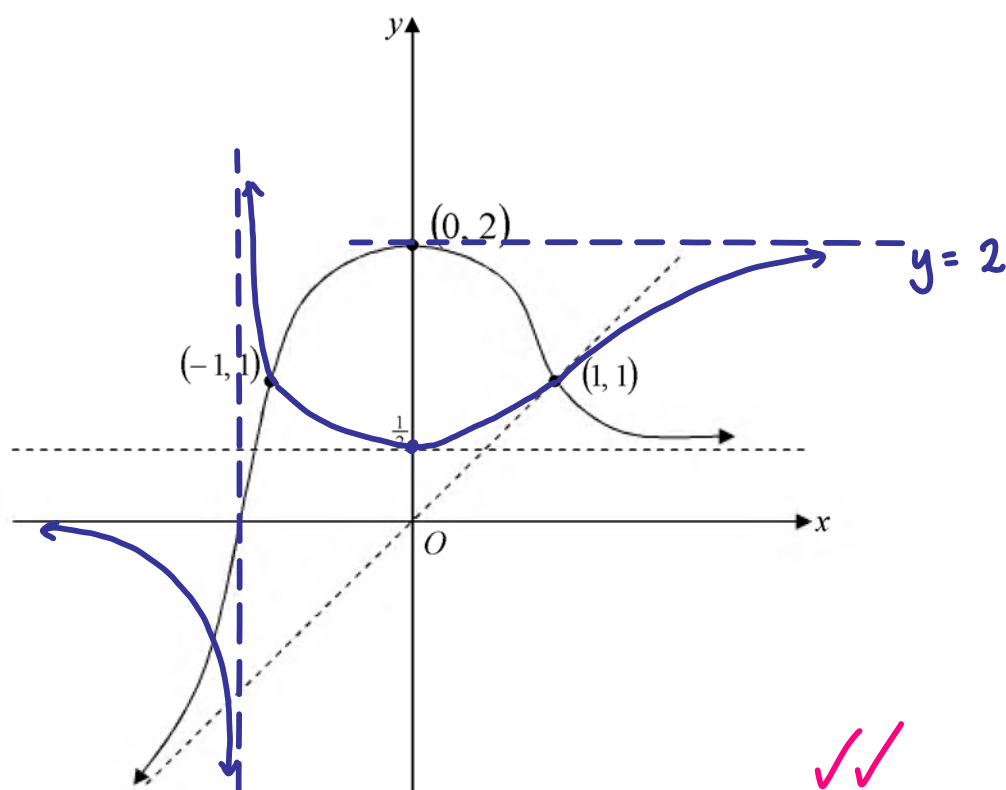
(i)

$$y = (f(x))^2$$



(ii)

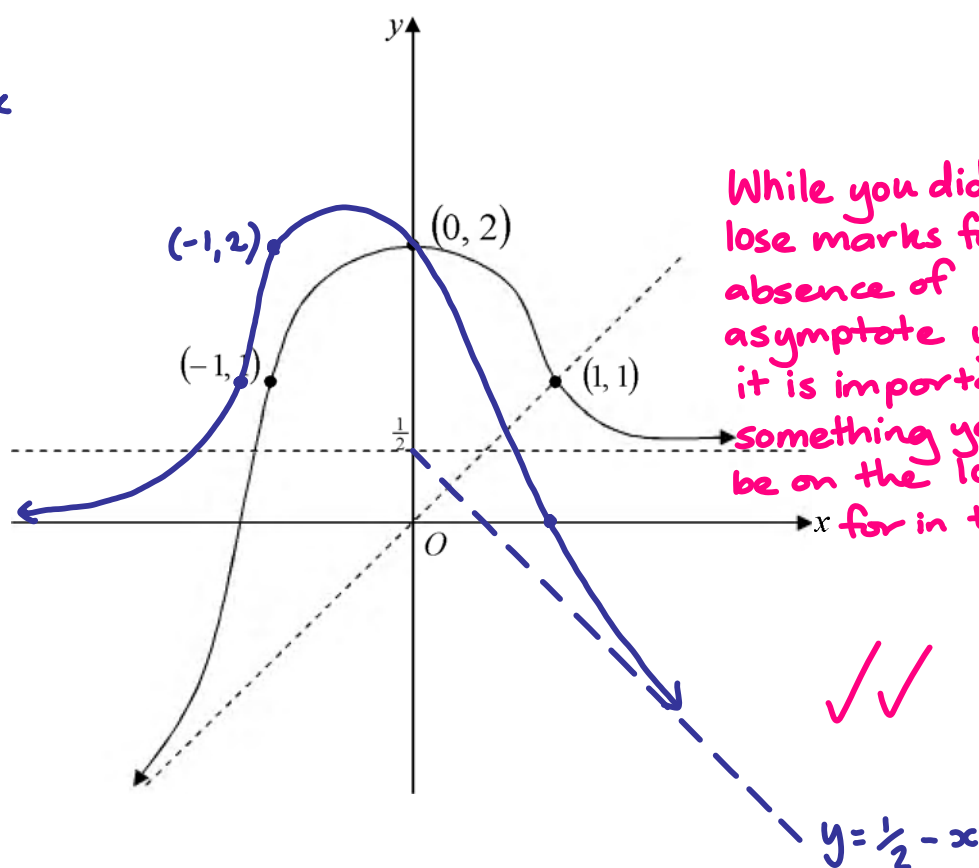
$$y = \frac{1}{f(x)}$$



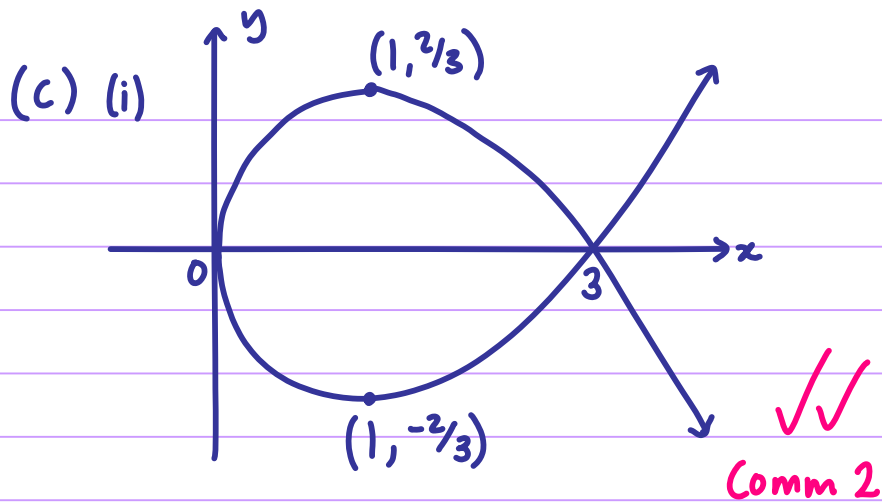
Question 3b (continued)

(iii)

$$y = f(x) - x$$



Comm 6



Important features:

- the vertical asymptote at the origin
- the curve intersects itself with a non-zero gradient at $(3,0)$
- max/min at $(1, \pm 2/3)$

(ii)

$$y^2 = \frac{1}{9} x(x-3)^2$$

$$= \frac{1}{9} (x^3 - 6x^2 + 9x)$$

$$2y \frac{dy}{dx} = \frac{1}{9} (3x^2 - 12x + 9)$$

$$= \frac{1}{3} (x^2 - 4x + 3)$$

$$\frac{dy}{dx} = \frac{(x-1)(x-3)}{6y} \quad \checkmark \text{ Calc 1}$$

(iii) length of the top half

$$= \int_0^3 \sqrt{1 + \frac{(x-1)^2(x-3)^2}{36y^2}} dx$$

$$= \int_0^3 \sqrt{1 + \frac{(x-1)^2(x-3)^2}{4x(x-3)^2}} dx \quad \checkmark$$

$$= \int_0^3 \sqrt{\frac{4x + x^2 - 2x + 1}{4x}} dx$$

$$= \int_0^3 \frac{x+1}{2\sqrt{x}} dx \quad \checkmark$$

$$= \left[\frac{x^{3/2}}{3} + x^{1/2} \right]_0^3$$

$$= 2\sqrt{3} \quad \checkmark \quad \text{Calc 3}$$

$\therefore$ total length of the curve is $4\sqrt{3}$ units.

Not many had the faith that the algebra/integral would work out. Sometimes you just need to be confident in your own algebra & believe that eventually the integral will be doable.

Full marks were given for an answer of $2\sqrt{3}$, but in fact to get the total length you need to double this answer.

Question 4 (15 marks)

Calc 6

(a) $P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$

(i) Since $P(x)$ has real coefficients, if $a+bi$ & $a+2bi$ are roots, the conjugates $a-bi$ & $a-2bi$ are also roots.

You must state this theorem whenever you need to use it.

Sum of roots = $-b/a$

$$a+bi + a-bi + a+2bi + a-2bi = 4$$

$$4a = 4$$

$$a = 1$$

Product of roots = $+c/a$

$$(a+bi)(a-bi)(a+2bi)(a-2bi) = 10$$

$$(a^2 + b^2)(a^2 + 4b^2) = 10$$

$$(1 + b^2)(1 + 4b^2) = 10$$

$$4b^4 + 5b^2 - 9 = 0$$

$$(4b^2 + 9)(b^2 - 1) = 0$$

$$b = 1 \quad \text{since } b \in \mathbb{R} \text{ \& } b > 0$$

you really do need to state exactly why $b \neq \pm\sqrt{-\frac{9}{4}}, -1$

(ii) The roots are: $1 \pm i, 1 \pm 2i$

$$1+i, 1-i \rightarrow \text{sum} = 2, \text{prod} = 2$$

$$1+2i, 1-2i \rightarrow \text{sum} = 2, \text{prod} = 5$$

$$\therefore P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$$

$$= (x^2 - 2x + 2)(x^2 - 2x + 5)$$

$$\begin{aligned}
 (b) \quad V &= \int_0^{\pi/2} 2\pi x \cos x \, dx \quad \checkmark \\
 &= 2\pi \left(\left[x \cdot \sin x \right]_0^{\pi/2} - \int_0^{\pi/2} \sin x \, dx \right) \quad \checkmark \\
 &= 2\pi \left(\frac{\pi}{2} - 0 + \left[\cos x \right]_0^{\pi/2} \right) \\
 &= 2\pi \left(\frac{\pi}{2} - 1 \right) \\
 &= \pi^2 - 2\pi \text{ units}^3. \quad \checkmark \quad \text{Calc 3}
 \end{aligned}$$

(c) Taking slices perpendicular to the axis:

$$\begin{aligned}
 V &= 2\pi \int_0^{a\sqrt{3}} x^2 - a^2 \, dy \quad \checkmark \\
 &= 2\pi \int_0^{a\sqrt{3}} 4a^2 - y^2 - a^2 \, dy \\
 &= 2\pi \left[3a^2 y - \frac{y^3}{3} \right]_0^{a\sqrt{3}} \quad \checkmark \\
 &= 2\pi \left[3a^3\sqrt{3} - a^3\sqrt{3} - 0 \right] \\
 &= 4\pi a^3\sqrt{3} \text{ cm}^3. \quad \checkmark
 \end{aligned}$$

Calc 3

This question could have been done by shells as well.

Too many got caught up in wanting to subtract a volume from $\frac{4}{3}\pi(2a)^3$ - which just made it harder.

$$(d) \quad x^3 + kx + 1 = 0$$

$$(i) \quad S_1 = \alpha + \beta + \gamma \\ = 0 \quad \checkmark$$

$$S_2 = \alpha^2 + \beta^2 + \gamma^2 \\ = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma) \\ = 0^2 - 2 \times k \\ = -2k \quad \checkmark$$

$$(ii) \quad S_{n+3} + kS_{n+1} + S_n$$

Not everything is induction!

$$= (\alpha^{n+3} + \beta^{n+3} + \gamma^{n+3}) \\ + k(\alpha^{n+1} + \beta^{n+1} + \gamma^{n+1}) \\ + (\alpha^n + \beta^n + \gamma^n) \quad \checkmark \\ = \alpha^n(\alpha^3 + k\alpha + 1) + \beta^n(\beta^3 + k\beta + 1) \\ + \gamma^n(\gamma^3 + k\gamma + 1)$$

$$= \alpha^n \times 0 + \beta^n \times 0 + \gamma^n \times 0$$

$$= 0 \quad (\text{since } \alpha, \beta, \gamma \text{ are roots} \quad \checkmark \\ \text{\& satisfy the equation})$$

$$(iii) \quad S_4 + kS_2 + S_1 = 0$$

$$S_4 = -S_1 - kS_2 \\ = -0 - k \times -2k \\ = 2k^2$$

$$\therefore \alpha^4 + \beta^4 + \gamma^4 = 2k^2 \quad \checkmark$$

Question 5 (15 marks)

Calc 6, Reas 2

$$(a)(i) \quad P: \begin{aligned} x &= a \sec \theta & y &= b \tan \theta \\ \frac{dx}{d\theta} &= a \sec \theta \tan \theta & \frac{dy}{d\theta} &= b \sec^2 \theta \end{aligned}$$

$$\Rightarrow \frac{dy}{dx} = \frac{b \sec^2 \theta}{a \sec \theta \tan \theta} = \frac{b \sec \theta}{a \tan \theta}$$

$$\Rightarrow m_N = -\frac{a \tan \theta}{b \sec \theta} \quad \checkmark$$

$\therefore$ EQN OF NORMAL:

$$y - b \tan \theta = -\frac{a \tan \theta}{b \sec \theta} (x - a \sec \theta)$$

$$\frac{by}{\tan \theta} - b^2 = -\frac{ax}{\sec \theta} + a^2$$

$$\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2 \quad \checkmark$$

Calc 2

$$(ii) \quad A: \left(\frac{(a^2 + b^2) \sec \theta}{a}, 0 \right) \quad \checkmark$$

$$B: \left(0, \frac{(a^2 + b^2) \tan \theta}{b} \right) \quad \checkmark$$

$$\therefore M = \left(\frac{(a^2 + b^2) \sec \theta}{2a}, \frac{(a^2 + b^2) \tan \theta}{2b} \right) \quad \checkmark$$

$$(iii) \quad G = \left(\frac{(a^2 + b^2) \sec \theta}{3a}, \frac{(a^2 + b^2) \tan \theta}{3b} \right)$$

$\checkmark$ Part (iii) was an easy application of a formula & there was an abnormally high number who couldn't get it right.

(iv) Locus of G:

$$x = \frac{(a^2+b^2)\sec\theta}{3a} \Rightarrow \sec\theta = \frac{3ax}{a^2+b^2}$$

$$y = \frac{(a^2+b^2)\tan\theta}{3b} \Rightarrow \tan\theta = \frac{3by}{a^2+b^2}$$

To find the cartesian equation you need to get rid of the parameter.

$$\text{Since } \sec^2\theta - \tan^2\theta = 1$$

$$\frac{9a^2x^2}{(a^2+b^2)^2} - \frac{9b^2y^2}{(a^2+b^2)^2} = 1 \quad \checkmark$$

$$\frac{x^2}{\left(\frac{a^2+b^2}{3a}\right)^2} - \frac{y^2}{\left(\frac{a^2+b^2}{3b}\right)^2} = 1$$

Which is a hyperbola which intersects the x axis at

$$L = \left(\frac{a^2+b^2}{3a}, 0 \right) \quad \checkmark$$

$$(v) \frac{OL}{OK} < 1 \Rightarrow \frac{\frac{a^2+b^2}{3a}}{a} < 1$$

$$\Rightarrow a^2+b^2 < 3a^2$$

$$\Rightarrow b^2 < 2a^2$$

$$\Rightarrow a^2(e^2-1) < 2a^2$$

$$\Rightarrow e^2-1 < 2$$

$$\Rightarrow e^2 < 3 \quad \checkmark$$

$$\Rightarrow e < \sqrt{3} \quad (e > 0)$$

Don't be scared of inequalities. Just start with what you're given & aim to get to the end.

Also, since it's a hyperbola $e > 1$

$$[\text{or } b^2 = a^2(e^2-1) \Rightarrow e^2 = \frac{b^2}{a^2} + 1 > 1 \Rightarrow e > 1]$$

$$\therefore 1 < e < \sqrt{3} \quad \checkmark$$

Reas 2

$$\begin{aligned}
 \text{(b) (i) Area} &= \frac{1}{2} (y+1)^2 \sin \frac{\pi}{3} \\
 &= \frac{1}{2} (\sec x + 1)^2 \times \frac{\sqrt{3}}{2} \\
 &= \frac{\sqrt{3}}{4} (\sec x + 1)^2. \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } V &= \int_0^{\pi/4} \frac{\sqrt{3}}{4} (\sec x + 1)^2 dx \\
 &= \frac{\sqrt{3}}{4} \int_0^{\pi/4} \sec^2 x + 2\sec x + 1 dx \\
 &= \frac{\sqrt{3}}{4} \left[\tan x + 2\ln(\sec x + \tan x) + x \right]_0^{\pi/4} \\
 &= \frac{\sqrt{3}}{4} \left[1 + 2\ln(\sqrt{2} + 1) + \frac{\pi}{4} - 0 \right] \\
 &= \frac{\sqrt{3}}{4} \left[1 + 2\ln(\sqrt{2} + 1) + \frac{\pi}{4} \right] \checkmark \text{ units}^3
 \end{aligned}$$

Calc 4

Question 6 (15 marks)

Calc 4, Reas 11

(a) (i) let $x = \pi - u$ $x = 0, u = \pi$
 $dx = -du$ $x = \pi, u = 0$

When doing a substitution you need to substitute EVERYTHING — the limits & dx too!

$$\begin{aligned} I_1 &= \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx \\ &= \int_{\pi}^0 \frac{(\pi - u) \sin(\pi - u)}{1 + \cos^2(\pi - u)} \cdot -du \\ &= \int_0^{\pi} \frac{(\pi - u) \sin u}{1 + \cos^2 u} du \quad \checkmark \\ &= \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx = I_2 \end{aligned}$$

(ii) $I_1 + I_2 = \int_0^{\pi} \frac{x \sin x + (\pi - x) \sin x}{1 + \cos^2 x} dx$

← Not many recognised this reverse chain rule.

$$\begin{aligned} &= -\pi \int_0^{\pi} \frac{(-\sin x)}{1 + \cos^2 x} dx \\ &= -\pi \left[\tan^{-1}(\cos x) \right]_0^{\pi} \quad \checkmark \end{aligned}$$

$$= -\pi \left(\tan^{-1}(-1) - \tan^{-1}(1) \right)$$

$$= -\pi \left(-\frac{\pi}{4} - \frac{\pi}{4} \right)$$

$$= \frac{\pi^2}{2} \quad \checkmark$$

$$\therefore I_1 = I_2 = \frac{\pi^2}{4}$$

$$\therefore \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi^2}{4} \quad \checkmark$$

Calc 4

$$\begin{aligned}
 (b) (i) \frac{z^2-1}{z} &= z - z^{-1} \\
 &= \text{cis } \theta - \text{cis}(-\theta) \\
 &= \cos \theta + i \sin \theta - (\cos(-\theta) + i \sin(-\theta)) \\
 &= \cos \theta + i \sin \theta - \cos \theta - -i \sin \theta \\
 &= 2i \sin \theta
 \end{aligned}$$

OR/

$$\begin{aligned}
 \frac{z^2-1}{z} &= \frac{(\cos \theta + i \sin \theta)^2 - 1}{\cos \theta + i \sin \theta} \times \frac{\cos \theta - i \sin \theta}{\cos \theta - i \sin \theta} \\
 &= \frac{(\cos^2 \theta + \sin^2 \theta)(\cos \theta + i \sin \theta) - (\cos \theta - i \sin \theta)}{\cos^2 \theta + \sin^2 \theta} \\
 &= \cos \theta + i \sin \theta - \cos \theta + i \sin \theta \\
 &= 2i \sin \theta
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad z + z^3 + z^5 + z^7 + z^9 \\
 &= \frac{z((z^2)^5 - 1)}{z^2 - 1} \\
 &= \frac{z^{10} - 1}{\frac{z^2 - 1}{z}} \\
 &= \frac{\cos 10\theta + i \sin 10\theta - 1}{2i \sin \theta} \times \frac{-i}{-i} \\
 &= \frac{\sin 10\theta + i(1 - \cos 10\theta)}{2 \sin \theta}
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad \text{Re}(z + z^3 + z^5 + z^7 + z^9) \\
 &= \text{Re}\left(\frac{\sin 10\theta + i(1 - \cos 10\theta)}{2 \sin \theta}\right)
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \cos \theta + \cos 3\theta + \cos 5\theta + \cos 7\theta + \cos 9\theta \\
 &= \frac{\sin 10\theta}{2 \sin \theta}
 \end{aligned}$$

Each part of this question was very clear & followed directly from the previous part. You need to be confident following the lead & direction given in such questions.

$$\cos \theta + \cos 3\theta + \cos 5\theta + \cos 7\theta + \cos 9\theta = \frac{1}{2}$$

$$\Rightarrow \frac{\sin 10\theta}{2\sin \theta} = \frac{1}{2}$$

$$\Rightarrow \sin 10\theta = \sin \theta$$

$$10\theta = \theta + 2\pi k, \quad 10\theta = \pi - \theta + 2\pi k$$

$$9\theta = 2\pi k, \quad 11\theta = \pi + 2\pi k$$

$$\theta = \frac{2\pi k}{9}, \quad \frac{\pi + 2\pi k}{11} \quad k \in \mathbb{Z} \quad \checkmark \checkmark$$

Reas 7

(c) (i)

ways to choose the pairs of players for each of the 3 matches

$$\frac{\binom{7}{2} \binom{5}{2} \binom{3}{2}}{3!} \times \binom{1}{1} = 105$$

overcounted by 3! last player has a bye

since choosing A v B, C v D, E v F is equivalent to A v B, E v F, C v D etc.

(ii) Case 1: Amy or Ben has a bye

choose who has a bye $\rightarrow 2 \times \frac{\binom{6}{2} \binom{4}{2} \binom{2}{2}}{3!} = 30$

Case 2: Amy & Ben both play

Amy's opponent $\rightarrow 5 \times$ Ben's opponent $\rightarrow 4 \times$ other 2 play $\times \frac{\binom{3}{2}}{2!} \times \binom{1}{1} = 60$
person with bye

$$\therefore \text{Total \# ways} = 90 \quad \checkmark \checkmark$$

Reas 4

These type of General Solution questions are absolutely standard & there is no excuse for not knowing this work.

This perms & combs question was quite successful for those who attempted it. Don't be terrified of perms & combs - they are very doable & practice does make perfect.

OR, # ways Amy plays Ben

$$= \frac{\binom{5}{2} \binom{3}{2}}{2!} \times \binom{1}{1} = 15$$

$\therefore$ # ways Amy is not playing Ben = $105 - 15 = 90 \quad \checkmark \checkmark$

Question 7 (15 marks)

Calc 6, Reas 9

(a) (i) $(x-y)^2 \geq 0$

$$x^2 + y^2 - 2xy \geq 0$$

$$x^2 + y^2 \geq 2xy \quad \checkmark$$

(ii) Dividing by xy

$$\Rightarrow \frac{x}{y} + \frac{y}{x} \geq 2 \quad (\text{since } x, y > 0) \quad \checkmark$$

$$\begin{aligned} \text{(iii)} \quad x^3 + y^3 &= (x+y)(x^2 - xy + y^2) \\ &\geq (x+y)(2xy - xy) \\ &= xy(x+y) \\ &= xyz \left(\frac{x}{z} + \frac{y}{z} \right) \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad \text{Similarly, } y^3 + z^3 &\geq xyz \left(\frac{y}{x} + \frac{z}{x} \right) \\ &\& z^3 + x^3 \geq xyz \left(\frac{z}{y} + \frac{x}{y} \right) \end{aligned}$$

Adding these:

$$\begin{aligned} 2(x^3 + y^3 + z^3) &\geq xyz \left(\frac{x}{z} + \frac{y}{z} + \frac{y}{x} + \frac{z}{x} + \frac{z}{y} + \frac{x}{y} \right) \\ &\geq xyz(2+2+2) = 6xyz \end{aligned}$$

$$\therefore x^3 + y^3 + z^3 \geq 3xyz \quad \checkmark$$

(v) $x = \sqrt[3]{a}, y = \sqrt[3]{b}, z = \sqrt[3]{c} \Rightarrow a+b+c \geq 3\sqrt[3]{abc}$

Similarly $a+b+d \geq 3\sqrt[3]{abd}$

$$a+c+d \geq 3\sqrt[3]{acd}$$

$$b+c+d \geq 3\sqrt[3]{bcd}$$

Multiplying these

$$\Rightarrow (a+b+c)(a+b+d)(a+c+d)(b+c+d) \geq 81abcd \quad \checkmark$$

Reas 5

Remember the technique of applying an inequality several times over & then putting it together.

(b) (i) A divides CB externally in the ratio $CA:AB = \varphi+1:\varphi$

B divides AC internally in the ratio $AB:BC = \varphi:1$

$$\therefore \frac{\varphi+1}{\varphi} = \frac{\varphi}{1}$$

$$\varphi^2 = \varphi+1$$



(ii) Solutions to $\varphi^2 = \varphi+1$

$$\varphi^2 - \varphi - 1 = 0$$

$$\varphi = \frac{1 \pm \sqrt{5}}{2}$$

$$\therefore \varphi = \frac{1+\sqrt{5}}{2}, \quad \theta = \frac{1-\sqrt{5}}{2}$$

When $n=1$: LHS = $F_1 = 1$

$$\text{RHS} = \frac{\varphi - \theta}{\sqrt{5}} = \frac{\sqrt{5}}{\sqrt{5}} = 1$$

& when $n=2$: LHS = $F_2 = 1$

$$\text{RHS} = \frac{\varphi^2 - \theta^2}{\sqrt{5}}$$

$$= \frac{(\varphi+1) - (\theta+1)}{\sqrt{5}}$$

$$= \frac{\varphi - \theta}{\sqrt{5}} = \frac{\sqrt{5}}{\sqrt{5}} = 1$$



It's true for $n=1$ & $n=2$ so let $k-1$ & k be integers for which it's true i.e. $F_{k-1} = \frac{\varphi^{k-1} - \theta^{k-1}}{\sqrt{5}}$

$$\& F_k = \frac{\varphi^k - \theta^k}{\sqrt{5}}$$

I really liked this question but not many gave it a go which is a pity.

$$\begin{aligned}
\text{Then } F_{k+1} &= F_k + F_{k-1} \\
&= \frac{\varphi^k - \theta^k}{\sqrt{5}} + \frac{\varphi^{k-1} - \theta^{k-1}}{\sqrt{5}} \quad \checkmark \\
&\quad (\text{from assumption}) \\
&= \frac{\varphi^{k-1}(\varphi + 1) - \theta^{k-1}(\theta + 1)}{\sqrt{5}} \\
&= \frac{\varphi^{k-1} \cdot \varphi^2 - \theta^{k-1} \cdot \theta^2}{\sqrt{5}} \\
&\quad (\text{since } \varphi^2 = \varphi + 1 \text{ \& } \theta^2 = \theta + 1) \\
&= \frac{\varphi^{k+1} - \theta^{k+1}}{\sqrt{5}} \quad \checkmark
\end{aligned}$$

& so it's true for the next integer $k+1$

$\therefore$ By strong induction, $F_n = \frac{\varphi^n - \theta^{n-1}}{\sqrt{5}}$

(c) $\uparrow\uparrow$ $\downarrow g + kv$ $t=0$
 $v=u$
 $\ddot{x} = -g - kv$

This standard question was well done by most.

(i) $\frac{dv}{dt} = -(g + kv)$

$$\int_u^0 \frac{dv}{g + kv} = \int_0^T -dt$$

$$\left[\frac{1}{k} \ln(g + kv) \right]_u^0 = [-t]_0^T \quad \checkmark \checkmark$$

$$\frac{1}{k} \ln g - \frac{1}{k} \ln(g + ku) = -T + 0$$

$$T = \frac{1}{k} \ln \left(\frac{g + ku}{g} \right)$$

$$T = \frac{1}{k} \ln \left(1 + \frac{ku}{g} \right) \quad \checkmark$$

(2nd $\checkmark$ for finding C if an indefinite integral was done, then 3rd $\checkmark$ for $v=0$ & successfully finding T)

(ii) $v \frac{dv}{dx} = -(g + kv)$

$$\int_u^0 \frac{v dv}{g + kv} = \int_0^H -dx$$

$$\frac{1}{k} \int_u^0 \frac{g + kv}{g + kv} - \frac{g}{g + kv} dv = \int_0^H -dx$$

$$\frac{1}{k} \left[v - \frac{g}{k} \ln(g + kv) \right]_u^0 = [-x]_0^H \quad \checkmark \checkmark$$

$$\frac{1}{k} \left(0 - \frac{g}{k} \ln g - u + \frac{g}{k} \ln(g + ku) \right) = -H$$

$$H = \frac{1}{k} \left(u - g \left(\frac{1}{k} \ln(g + ku) - \frac{1}{k} \ln g \right) \right)$$

$$= \frac{1}{k} \left(u - g \cdot \frac{1}{k} \ln \left(1 + \frac{ku}{g} \right) \right) = \frac{u - gT}{k} \quad \checkmark$$

(2nd $\checkmark$ for finding C if an indefinite integral was done, then 3rd $\checkmark$ for $v=0$ & successfully finding H)

Question 8 (15 marks)

Reas 15

(a)(i) If α is a double root of $x^n - bx^2 + c = 0$
then $\alpha^n - b\alpha^2 + c = 0$ ①

& α is also a single root of $nx^{n-1} - 2bx = 0$
 $\Rightarrow n\alpha^{n-1} - 2b\alpha = 0$ ② ✓

$$\textcircled{2} \times \alpha \Rightarrow n\alpha^n - 2b\alpha^2 = 0$$
$$\alpha^n = \frac{2b\alpha^2}{n} \quad \textcircled{3}$$

substitute into ①

$$\Rightarrow \frac{2b\alpha^2}{n} - b\alpha^2 + c = 0$$

$$\alpha^2 \left(\frac{2b - nb}{n} \right) = -c$$

$$\alpha^2 = \frac{nc}{nb - 2b} \quad \checkmark$$

(ii) Substituting this into ③

$$\Rightarrow \left(\frac{nc}{nb - 2b} \right)^{n/2} = \frac{2b}{n} \cdot \frac{nc}{nb - 2b} \quad \checkmark$$

$$\Rightarrow n^{n/2} c^{n/2} \cdot (n-2) = 2c \cdot (n-2)^{n/2} b^{n/2}$$

squaring both sides

$$\Rightarrow n^n c^n (n-2)^2 = 4c^2 (n-2)^n b^n$$

$$\Rightarrow n^n c^{n-2} = 4b^n (n-2)^{n-2} \quad \checkmark$$

Many got an easy first mark - but some didn't because of trivial differentiating errors which is a pity! You've got to be able to maintain accuracy even in a rush.

The rest of this question was barely attempted.

(b) (i) $AB \parallel DM$ (both lines are $\perp$ to AC)
 $\therefore BM = MC \Rightarrow AD = DC$

(intercepts on parallel lines
 AB & DM are in the same ratio)

$\therefore PM$ bisects AC at right angles
& hence PM is the diameter
(the perpendicular bisector of a
chord passes through the centre)

(ii) In $\triangle MCD$ & $\triangle MPC$

$$\angle MDC = 90^\circ = \angle MCP$$

(given $MD \perp AC$ & angles
in a semicircle are 90°)

$$\angle CMD = \angle PMC \text{ (common)}$$

$\therefore \triangle MCD \parallel \triangle MPC$ (AA similarity test)

$$(iii) \frac{MD}{MC} = \frac{MC}{MP} \text{ (corresponding sides in } \parallel \text{ } \Delta\text{s in the same ratio)}$$

$$\Rightarrow \frac{MD}{MB} = \frac{MB}{MP} \text{ (since } MC = MB)$$

In $\triangle DMB$ & $\triangle BMP$

$$\frac{MD}{MB} = \frac{MB}{MP} \text{ (above)}$$

$$\angle DMB = \angle BMP \text{ (common)}$$

$\therefore \triangle DMB \parallel \triangle BMP$ (SAS similarity test)

Well done to those who
recognised this was an
easy 2 marks to pick up.

(iv) $\angle DBM = \angle BPM$ (corresponding angles in $\parallel$ Δ s are equal)

$\angle BPM = \angle ABK$ (alternate angles on parallel lines AB & PM)

$$\therefore \angle DBM = \angle ABK$$

(v) In ΔABD & ΔHBK

$$\angle DAB = \angle KHB = 90^\circ \text{ (given)}$$

$$\angle ABD = \angle ABC - \angle DBC$$

$$= \angle HBA - \angle KBA$$

($\angle ABC = \angle HBA$ common

& $\angle DBC = \angle KBA$ part iv)

$$= \angle HBK$$

$\therefore \Delta ABD \parallel \Delta HBK$ (AA similarity test)

In ΔDCB & ΔKAB

$$\angle CBD = \angle ABK \text{ (part iv)}$$

$$\angle DCB = 180 - 90 - \angle ABC \text{ (L sum } \Delta = 180^\circ)$$

$$= 180 - 90 - \angle HBA \text{ (common)}$$

$$= \angle KAB \text{ (L sum } \Delta = 180^\circ)$$

$\therefore \Delta DCB \parallel \Delta KAB$ (AA similarity test)

$$\frac{AD}{HK} = \frac{DB}{KB} \text{ (corresponding sides in } \parallel \Delta \text{s ABD \& HBK in same ratio)}$$

$$\frac{DB}{KB} = \frac{DC}{KA} \text{ (corresponding sides in } \parallel \Delta \text{s DCB \& KAB in same ratio)}$$

$$\therefore \frac{AD}{HK} = \frac{DC}{KA} \Rightarrow \frac{AD}{DC} = \frac{HK}{KA}$$

$$\Rightarrow \therefore \frac{HK}{KA} = 1 \text{ (since } AD=DC)$$

$$\therefore HK = KA$$

